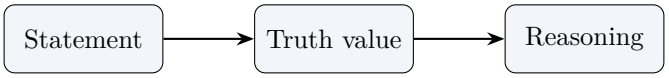
DISCRETE STRUCTURES

CMSC250

Chapter 1: Introduction to Discrete Mathematics

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Statements, truth values, and the language of logical reasoning



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1 Why do we study discrete mathematics?

In calculus, we often study quantities that vary continuously: the speed of a moving object, the slope of a curve, or the area under a graph. *Discrete mathematics* instead studies mathematical objects that can be distinguished or counted individually, together with the logical relationships between them. Examples include integers, graphs, sets, logical statements, and finite sequences.

This is especially useful in computer science. A program makes decisions based on conditions that are true or false. A circuit combines binary signals. An algorithm must be shown correct for *every* allowed input, not merely for the examples on which we tested it. Discrete mathematics provides the vocabulary and proof techniques for making these claims precise.

Consider the claim, "If a user enters a correct password, the system grants access." It sounds straightforward, but it raises several questions. Does this also mean the system grants access *only* when the password is correct? Does the claim say anything about what happens when the password is wrong? How can we express the rule unambiguously? These are questions of logic, and logic is where CMSC250 begins.

Key idea. The goal of this course is not simply to calculate answers. It is to state claims precisely, decide what they mean, and justify why they are true or false.

2 What is a statement?

A **statement** (also called a *proposition*) is a declarative sentence that has a definite truth value: it is either **true** or **false**. We do not need to know which truth value it has in order for it to qualify as a statement.

For instance, "There is life in the Andromeda galaxy" is a statement. We may not know its truth value, but it describes something that is either true or false. By contrast, "Is there life in the Andromeda galaxy?" is a question, not a statement.

2.1 Statements versus non-statements

Sentence	Statement?	Reason
$2 + 3 = 5$	Yes	Has truth value true.
$2 + 3 = 8$	Yes	Has truth value false.
There is life in another galaxy.	Yes	Truth may be unknown, but is definite.
$2x + 4 = 18$	No	With x unspecified, its truth depends on x.
$2x + 4 = 18$ when $x = 7$.	Yes	The value is fixed, so the claim is true.
What is the answer?	No	A question, not a declaration.
Please close the door.	No	A command, not a truth-valued claim.
Books are more interesting than movies.	Generally no	As an unqualified opinion, it lacks a fixed objective truth value in the course's intended interpretation.

2.2 Why an unknown truth value is not the same as no truth value

Students sometimes reason that a sentence cannot be a statement unless we can verify it. That is not the definition. A statement must *have* a truth value, not necessarily a truth value known to us. A difficult mathematical conjecture is still a statement. An open-ended question is not.

A second common mistake is to assume that every sentence containing a variable is automatically a statement. The sentence $x + 2 = 5$ is an *open sentence* when x is unspecified. But after assigning $x = 3$, or after adding a quantifier such as "for every integer x," it becomes a statement. We will study this formally in the predicate-logic chapters.

Worked example: classifying borderline cases

Decide whether each sentence is a statement.

a) "The exterior of a particular building has exactly 592 panes of glass." **Yes.** It is objectively true or false even if counting the panes would take time.

b) "$24 = 42$." **Yes.** A false proposition is still a proposition.

c) "$x^2 = 9$." **No, as written.** The variable is not assigned a value or quantified.

d) "For every integer x, $x^2 \geq 0$." **Yes.** It is a complete claim with a truth value.

e) "Math is fun." **Not in the intended introductory-logic classification.** It expresses a subjective preference rather than a fixed mathematical proposition.

3 Truth values and interpretations

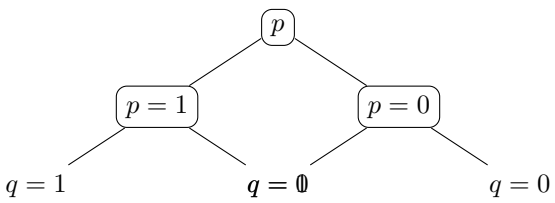
We use 1 for **true** and 0 for **false** in truth tables. This is the convention required in the supplied CMSC250 homework and graded assessments. Other books sometimes use T and F, but we will consistently use 1 and 0.

A **propositional variable** is a symbol such as p, q, or r standing for a statement. For example, let

$$p = \text{"The library is open,"} \qquad q = \text{"I have my student ID."}$$

An **interpretation** assigns a truth value to each propositional variable. One possible interpretation is $p = 1, q = 0$: the library is open, but I do not have my ID. Another is $p = 0, q = 1$.

For n independent propositional variables, there are 2^n possible interpretations because each variable has two choices. This explains why a complete truth table with three variables contains $2^3 = 8$ rows.



4 Negation: saying that a statement is false

The simplest logical operation is **negation**. If p is a statement, $\sim p$ means "not p." Negation reverses the truth value.

p	$\sim p$
1	0
0	1

Notation note. The symbols $\sim p$ and $\neg p$ both represent negation. In the CMSC250 materials supplied from Fawzi's class, $\sim$ is used more often. We will use $\sim$ in these notes, but you should recognize $\neg$ when reading other textbooks.

For example, if p means "The server is online," then $\sim p$ means "The server is not online." Negation does *not* mean "The server is broken." A server could be offline for maintenance. A correct negation denies exactly what the original statement asserts, without adding an unrelated explanation.

4.1 Double negation

Negating twice restores the original truth value:

$$\sim (\sim p) \equiv p.$$

Here $\equiv$ means that two expressions have the same truth value under every interpretation. We will formally develop logical equivalence in Chapter 3. For now, observe that "It is not true that the server is not online" says the same thing as "The server is online."

5 Combining statements: AND and OR

Logical connectives allow us to build compound statements from simpler ones. The two fundamental binary connectives are **conjunction** (AND), written $p \wedge q$, and **disjunction** (OR), written $p \vee q$.

5.1 Conjunction: $p \wedge q$

A conjunction is true only when *both* parts are true. Let p mean "I submitted the assignment" and q mean "I attached the required file." Then $p \wedge q$ says both events happened. If either one failed, the conjunction is false.

5.2 Disjunction: $p \vee q$

A disjunction is true when *at least one* part is true. Importantly, mathematical OR is normally **inclusive**: it also permits both statements to be true. If a course says "You may attend office hours on Tuesday or Thursday," the logical statement is still true if you attend both days, unless an exclusive interpretation is explicitly stated.

p	q	$p \wedge q$	$p \vee q$
1	1	1	1
1	0	0	1
0	1	0	1
0	0	0	0

Key idea. AND requires both claims. OR requires at least one. In mathematical logic, OR is inclusive unless the problem explicitly says otherwise.

6 Translating everyday language into logic

Translation is one of the most important early CMSC250 skills. The challenge is not memorizing symbols; it is identifying the exact logical conditions under which an English sentence would be true.

Suppose L means "I have a sock on my left foot" and R means "I have a sock on my right foot." Assume at most one sock can be worn on each foot.

English statement	Logical expression
I am wearing socks on both feet.	$L \wedge R$
I am wearing at least one sock.	$L \vee R$
I am not wearing any socks.	$\sim L \wedge \sim R$
I am wearing a sock on the left but not right.	$L \wedge \sim R$
I am wearing exactly one sock.	$(L \wedge \sim R) \vee (\sim L \wedge R)$
I am wearing at most one sock.	$\sim (L \wedge R)$

Why is "at most one" written as $\sim (L \wedge R)$? Because the only forbidden case is wearing *two* socks. The permitted cases are zero socks, left only, and right only. In contrast, "exactly one" forbids both zero and two.

Worked example: at least one versus exactly one

Let a mean "Alice attends" and b mean "Bob attends." Translate:

a) "At least one attends": $a \vee b$. Both may attend.

b) "Neither attends": $\sim a \wedge \sim b$.

c) "Exactly one attends": $(a \wedge \sim b) \vee (\sim a \wedge b)$.

d) "They do not both attend": $\sim (a \wedge b)$. This still permits neither to attend.

Notice how a single word—*at least*, *at most*, or *exactly*—changes the set of allowed interpretations.

7 How to build a truth table correctly

A truth table evaluates a compound statement under *every* possible interpretation. In Fawzi's CMSC250 course, the required layout is specific: variable columns are in alphabetical order, truth values use 1 and 0, and rows follow descending binary order.

For variables p, q, r, the eight rows must be

p	q	r
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

The pattern is systematic. The leftmost variable changes least frequently, while the rightmost changes every row. This is the same pattern as counting down in binary from 111_2 to 000_2. It guarantees that no interpretation is omitted or repeated.

Worked example: evaluate $(p \vee q) \wedge \sim p$

First list the four interpretations of p, q. Next compute $p \vee q$, then $\sim p$, and finally combine those intermediate columns with AND.

p	q	$p \vee q$	$\sim p$	$(p \vee q) \wedge \sim p$
1	1	1	0	0
1	0	1	0	0
0	1	1	1	1
0	0	0	1	0

The compound statement is true only in the third row, when p is false and q is true. In plain English, the formula says "at least one of p and q is true, but p is false." Therefore q must be true.

8 Common mistakes and grading pitfalls

1. **Confusing false with not a statement.** "$2 + 2 = 5$" is false, but it is still a statement.
2. **Treating OR as exclusive.** $p \vee q$ is true when both p and q are true.
3. **Using the wrong truth-table convention.** Use alphabetical variable columns and the required $1, 0$ pattern. The supplied homework warns that ignoring this convention can invalidate a truth-table answer on an assessment.
4. **Adding information when negating.** The negation of "The program terminated" is "The program did not terminate," not "The program crashed."

5. **Mixing up exactly one and at least one.** "Exactly one" requires excluding the case where both are true.

6. **Using = where logical equivalence is intended.** Later, when comparing formulas, use $\equiv$, not arithmetic equality, and justify equivalence carefully.

9 Practice problems

Try these before consulting the solutions. They are original exercises designed to reflect the introductory reasoning used in CMSC250.

1. Classify each as a statement or non-statement: (a) "5 is prime"; (b) "$y + 1 = 8$"; (c) "There are infinitely many primes"; (d) "Please write your answer clearly."

2. Let p mean "The file exists" and q mean "The file is readable." Translate: (a) both conditions hold; (b) neither holds; (c) at least one holds; (d) exactly one holds.

3. Construct a complete truth table for $\sim (p \wedge q)$ and $\sim p \vee \sim q$. What do you observe?

4. Let a, b, c mean three students attend a meeting. Write a formula meaning "at least one attends" and another meaning "none attends."

5. Explain why "I do not know whether p is true" does not imply that p is not a statement.

10 Solutions to practice problems

1. (a) Statement, true. (b) Not a statement as written, because y is unspecified. (c) Statement, true. (d) Not a statement: it is a command.

2. (a) $p \wedge q$. (b) $\sim p \wedge \sim q$. (c) $p \vee q$. (d) $(p \wedge \sim q) \vee (\sim p \wedge q)$.

3.

p	q	$\sim (p \wedge q)$	$\sim p \vee \sim q$
1	1	0	0
1	0	1	1
0	1	1	1
0	0	1	1

. The final columns match in every row, so the formulas are logically equivalent. This is one of De Morgan's laws, studied formally in Chapter 3.

4. At least one: $a \vee b \vee c$. None: $\sim a \wedge \sim b \wedge \sim c$.

5. Whether a sentence is a statement depends on whether it has a definite truth value, not on whether a person knows that value. For example, a factual claim about a distant planet remains true or false even if we have not investigated it.